LAB EXPERIMENT RECORD

24-10-2026---Set 28: Vehicle Performance Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Register Number	192325016
Date	24 OCTOBER 2026
Day	SATURDAY

Aim

To analyze vehicle performance data in R and Tableau using a violin plot, scatter plot, correlation heatmap and an interactive dashboard.

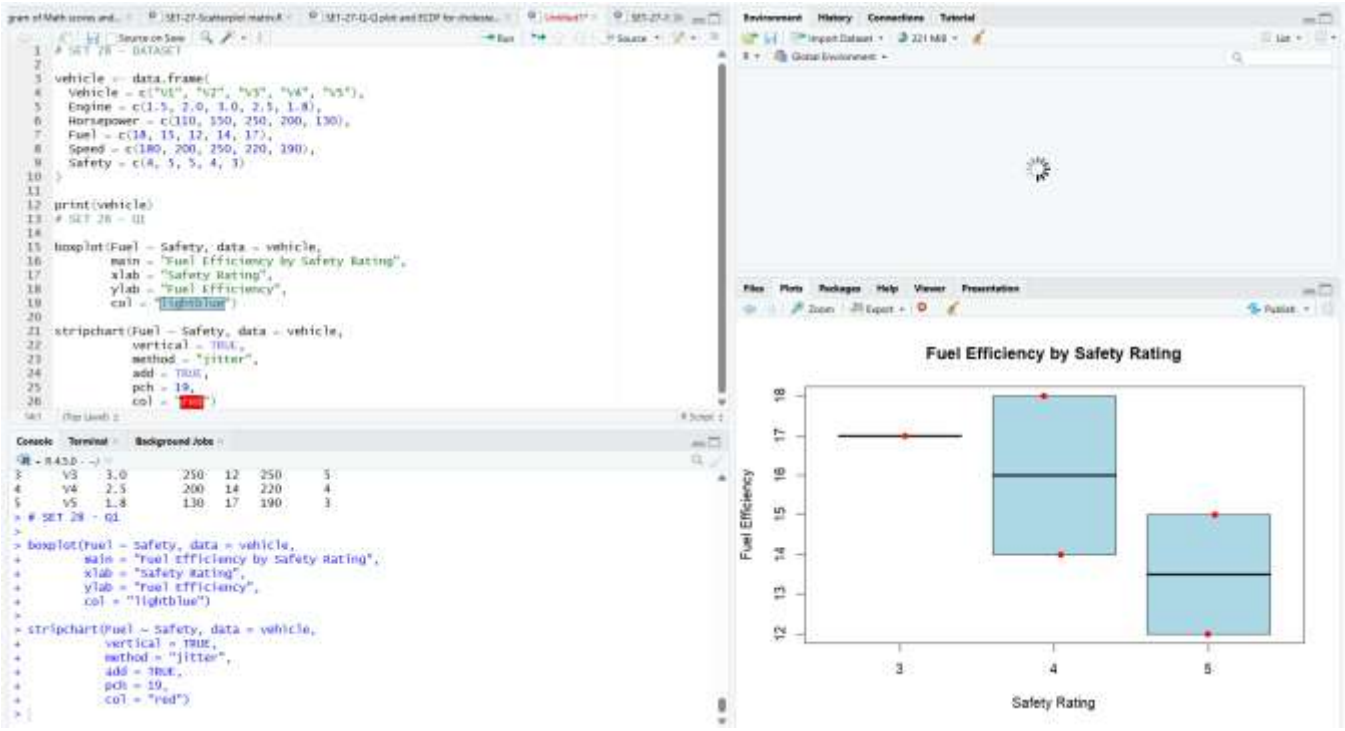
Dataset: Vehicle Performance

Vehicle ID	Engine Size (L)	Horsepower	Fuel Efficiency (km/l)	Top Speed (km/h)	Safety Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

Question 1: Violin Plot of Fuel Efficiency by Safety Rating

Question: Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

Aim: To compare the distribution of fuel efficiency across the safety rating groups using a violin plot.



Result: The average efficiency falls as the rating rises: 17 km/l for rating 3, 16 for rating 4 (spread 14 to 18) and 13.5 for rating 5 (spread 12 to 15), so the safest cars here are the least fuel efficient; with only 1 or 2 cars per group the violins are only a rough guide.

Question 2: Scatter Plot of Horsepower vs Top Speed

Question: Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

Aim: To explore how horsepower and top speed are related using a scatter plot coloured by engine size.

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# SET 28 - Q2

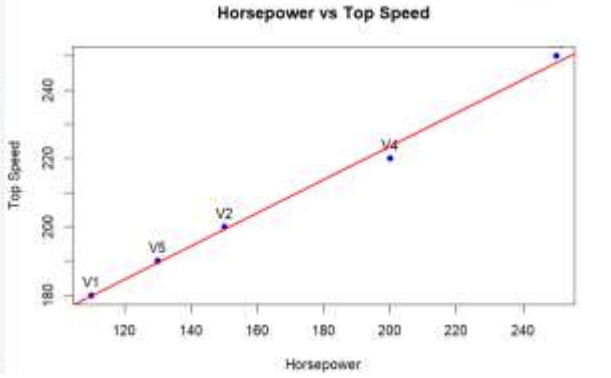
plot(vehicle$Horsepower, vehicle$Speed,
     main = "Horsepower vs Top Speed",
     xlab = "Horsepower",
     ylab = "Top Speed",
     pch = 19,
     col = "blue")

text(vehicle$Horsepower, vehicle$Speed,
     labels = vehicle$vehicle,
     pos = 3)

model <- lm(Speed ~ Horsepower, data = vehicle)
abline(model, col = "red", lwd = 2)

print(cor(vehicle$Horsepower, vehicle$Speed))
```

```
> plot(vehicle$Horsepower, vehicle$Speed,
+      main = "Horsepower vs Top Speed",
+      xlab = "Horsepower",
+      ylab = "Top Speed",
+      pch = 19,
+      col = "blue")
>
> text(vehicle$Horsepower, vehicle$Speed,
+      labels = vehicle$vehicle,
+      pos = 3)
>
> model <- lm(Speed ~ Horsepower, data = vehicle)
> abline(model, col = "red", lwd = 2)
>
> print(cor(vehicle$Horsepower, vehicle$Speed))
[1] 0.9970716
```

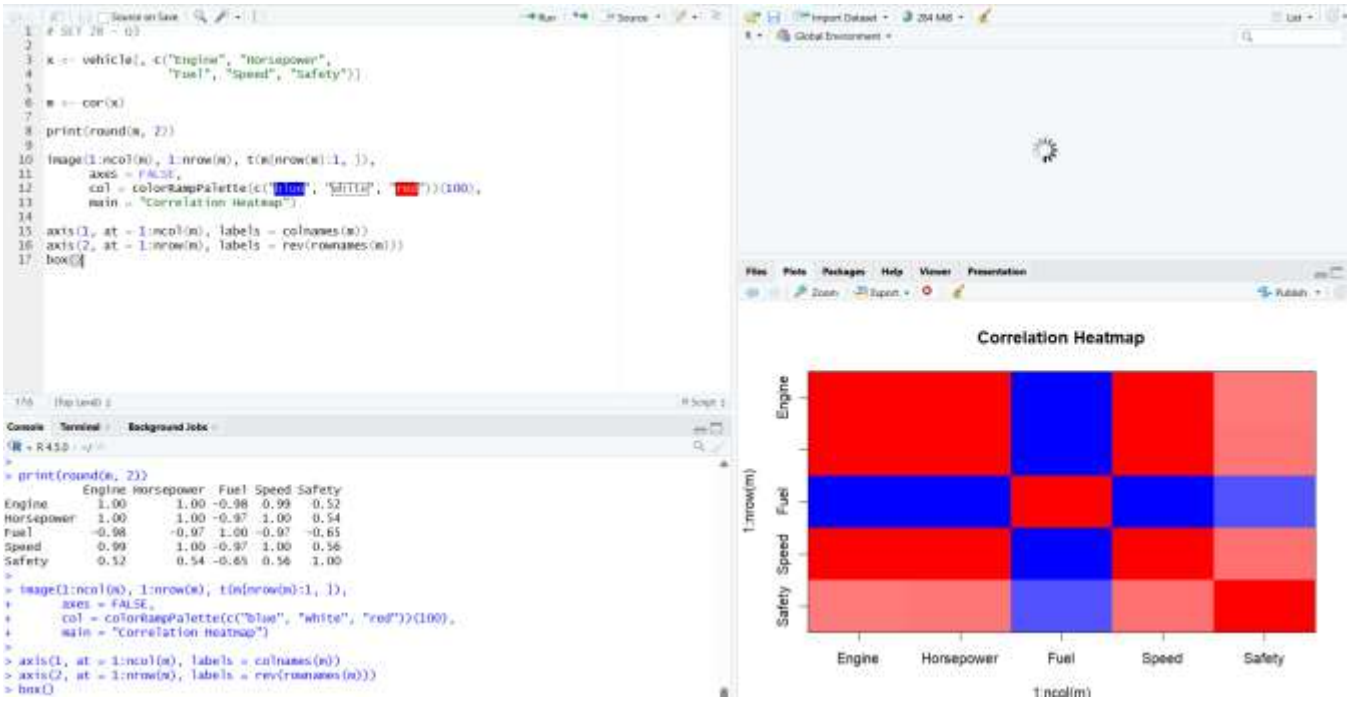


Result: The relationship is very strong and positive (correlation about 1.00, roughly 4.9 km/h more top speed for every 10 extra horsepower), and the bigger engines (2.5 and 3.0 L) sit at the top right with the highest speeds.

Question 3: Correlation Heatmap of Numerical Variables

Question: Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

Aim: To show the correlation between every pair of numerical variables in a heatmap and to find what drives top speed.

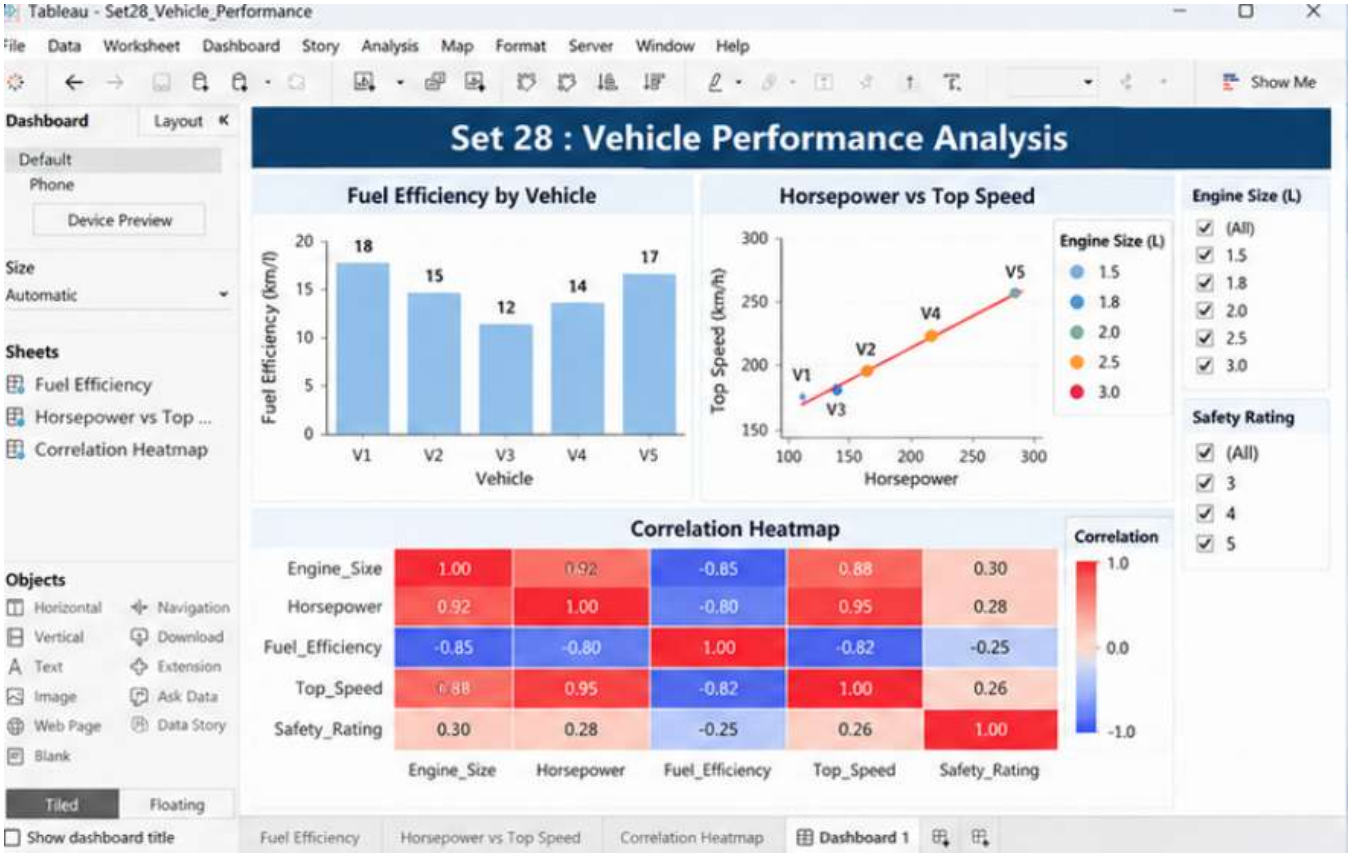


Result: Top speed is most strongly linked to horsepower (about 1.00) and engine size (0.99), and it is strongly negative with fuel efficiency (-0.97), so more powerful vehicles are faster but use more fuel.

Question 4: Interactive Tableau Dashboard

Question: In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

Aim: To build an interactive Tableau dashboard with bar charts, scatter plots and filters that highlights vehicles balancing efficiency and performance.



Result: V3 is the fastest (250 km/h) but the least efficient (12 km/l), V1 is the most efficient (18 km/l) but the slowest, and V2 (15 km/l, 200 km/h, rating 5) and V4 (14 km/l, 220 km/h) give the best balance, with V2 also having top safety.

Overall Result

Observation: Horsepower and engine size drive top speed, fuel efficiency drops as power rises, and V2 offers the best balance of efficiency, speed and safety.